

Jupyter Notebook Execution Report

Name: Group 6
Project Title: Lab1
Date: September 18, 2026

Cell 1: ■ Code

```
import numpy as np
```

Cell 2: ■ Code

```
q_3 = np.array([1, 2, 3, 4])  
q_3
```

Output:

```
array([1, 2, 3, 4])
```

Cell 3: ■ Code

```
q_4_ones = np.ones((3, 4))  
q_4_zeros = np.zeros((4, 3))  
q_4_ones  
q_4_zeros
```

Output:

```
array([[0., 0., 0.],  
       [0., 0., 0.],  
       [0., 0., 0.],  
       [0., 0., 0.]])
```

Cell 4: ■ Code

```
rng = np.random.default_rng()  
q_5_A = rng.random((2,3))  
q_5_B = rng.random((3,4))
```

```
np.matmul(q_5_A, q_5_B)
```

Output:

```
array([[1.04046527, 0.97308681, 1.3143886 , 0.93181335],  
       [0.94219441, 0.61096268, 0.82351033, 0.43254448]])
```

Cell 5: ■ Code

```
q_6 = np.array([[3,1], [1, 2]])  
eigvals, eigvecs = np.linalg.eig(q_6)
```

```
print(eigvals)  
print(eigvecs)
```

Output:

```
[3.61803399 1.38196601]  
[[ 0.85065081 -0.52573111]  
 [ 0.52573111  0.85065081]]
```